Physics Formula Sheet (College & Pre-Med)

Formulas, worked examples, clinical notes and exam traps for college physics, pre-med and MCAT review

MindShelf

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Units & conversions

1 atm = 101.325 kPa = 760 mmHg; 1 mmHg = 133.322 Pa ≈ 1.36 cmH_2O; 1 cmH_2O = 98.0665 Pa; 1 cal = 4.184 J; 1 Cal (food) = 1 kcal = 4,184 J; 1 eV = 1.602×10^{-19} J; T(K) = T(°C) + 273.15. 1 Pa = $1\,N/m^2$, 1 J = 1 N·m, 1 W = 1 J/s.

Kinematics

$$v = v_0 + at \quad \Delta x = v_0 t + \frac{1}{2}at^2 \quad v^2 = v_0^2 + 2a\Delta x$$

Projectile: v_x constant, $a_y = -g$; the vertical motion alone sets the flight time.

Forces & equilibrium

$\sum \vec{F} = m\vec{a}$; weight $F_g = mg$; spring $F = -kx$; friction $f_s \le \mu_s N$, $f_k = \mu_k N$. Incline: $F_{\parallel} = mg\sin\theta$, $N = mg\cos\theta$. Torque $\tau = rF\sin\theta$. Equilibrium: $\sum F = 0$ and $\sum \tau = 0$.

Work & energy

$W = Fd\cos\theta$; $K = \frac{1}{2}mv^2$; $U_g = mgh$; $U_s = \frac{1}{2}kx^2$; $W_{net} = \Delta K$; $P = W/t = Fv$. Without friction $K + U$ is conserved.

Momentum

$p = mv$; impulse $J = F\Delta t = \Delta p$. Momentum is conserved in every isolated collision, kinetic energy only in elastic ones. Perfectly inelastic: $m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$.

EXAMPLE Landing

70 kg at 5 m/s, $F = mv/\Delta t$: stopped in 0.5 s (knees bent) ⇒ 700 N; in 0.05 s (knees locked) ⇒ 7,000 N.

Circular motion & gravity

$$a_c = \frac{v^2}{r} = \omega^2 r \qquad F_c = \frac{mv^2}{r}$$

$$F = G\frac{m_1 m_2}{r^2} \qquad g = \frac{GM}{r^2}$$

Fluids & circulation

$\rho = m/V$; $P = F/A$; $P = P_0 + \rho gh$; buoyancy $F_B = \rho_{fluid} V_{sub} g$.

$Q = Av \quad A_1v_1 = A_2v_2 \quad P + \frac{1}{2}\rho v^2 + \rho gh = \text{const}$

$$Q = \frac{\pi r^4 \Delta P}{8\eta L} \qquad \Delta P = QR$$

Laplace: vessel $T = Pr$; alveolus $P = 2T/r$.

EXAMPLE Poiseuille and IV bag

Radius × k = 0.8 ⇒ flow × k^4 ≈ 41 % of baseline, resistance × 2.4. Bag 1 m above the cannula: ρgh = 1,000 × 9.8 × 1 Pa ≈ 74 mmHg.

Waves & sound

$v = f\lambda$; $T = 1/f$; SHM $T = 2\pi\sqrt{m/k}$, pendulum $T = 2\pi\sqrt{L/g}$.

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$$\beta = 10\log_{10}\frac{I}{I_0} \quad I = \frac{P}{4\pi r^2} \quad f' = f\frac{v \pm v_o}{v \mp v_s}$$

$I_0 = 1 \times 10^{-12}$ W/m². Doppler: upper signs when source and observer approach.

Light & optics

$n = c/v$; $n_1 \sin\theta_1 = n_2 \sin\theta_2$; critical angle $\sin\theta_c = n_2/n_1$.

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \quad m = -\frac{d_i}{d_o} \quad P = \frac{1}{f}$$

P in diopters (f in m). Mirror $f = R/2$. Lenses in contact: $P = P_1 + P_2$.

Sign	+	−
f	converging lens, concave mirror	diverging lens, convex mirror
d_i	real (lens: far side; mirror: front)	virtual
m	upright	inverted

EXAMPLE Lens and myopia
$f = 0.1$ m, $d_o = 0.3$ m ⇒ $d_i = 0.15$ m, $m = -0.5$ (real, inverted). Far point 0.5 m ⇒ lens $P = -1/x = -2$ D.

Electricity & magnetism

$$F = k\frac{q_1 q_2}{r^2} \quad E = \frac{kq}{r^2} \quad V = \frac{kq}{r} \quad B = \frac{\mu_0 I}{2\pi r}$$

$U = qV$; $\Delta V = Ed$; $F = qvB\sin\theta$; $F = ILB\sin\theta$ (right-hand rule).

Circuits

$V = IR$; $R = \rho L/A$; $P = IV = I^2R = V^2/R$. Series $R = R_1 + R_2$; parallel $1/R = 1/R_1 + 1/R_2$. $C = Q/V$; $C = \kappa\varepsilon_0 A/d$; $U = \frac{1}{2}CV^2$; capacitors combine the other way round. RC: $\tau = RC$, 63 % charged after one τ (37 % left on discharge). Kirchhoff: $\sum I_{in} = \sum I_{out}$; $\sum \Delta V = 0$ around a loop.

EXAMPLE Parallel and defibrillator
$6\,\Omega \parallel 3\,\Omega = 2\,\Omega$. $C = 0.0001$ F at 2,000 V: $U = \frac{1}{2}CV^2 = 200$ J.

Thermodynamics & gases

$PV = nRT$; $P_1V_1/T_1 = P_2V_2/T_2$; $\Delta U = Q - W$ (W by the gas); $W = P\Delta V$; $Q = mc\Delta T$; $Q = mL$; Dalton $P_i = x_i P_{total}$; mean $K = \frac{3}{2}k_B T$.

Atomic & nuclear

$E = hf = hc/\lambda$; $K_{max} = hf - \varphi$; $E = \Delta mc^2$; $A = \lambda N$; $\lambda = \ln 2/T_{\frac{1}{2}}$.

$$N = N_0\left(\frac{1}{2}\right)^{t/T_{\frac{1}{2}}}$$

α: $A - 4$, $Z - 2$; β^-: $Z + 1$; β^+: $Z - 1$; γ: no change. Dose D in Gy (J/kg); equivalent dose $H = w_R D$ in Sv.

EXAMPLE Photon and half-life
$E = hc/\lambda = 1{,}240$ eV·nm $/ 500$ nm $= 2.48$ eV. 800 MBq, 24 h, $T_{\frac{1}{2}} = 6$ h: $n = 4$ ⇒ 50 MBq.

CLINICAL NOTE

- Elbow: third-class lever, so the biceps pulls far harder than the load.
- Airbags and bent knees lengthen Δt: same Δp, lower force.
- $R \propto 1/r^4$: slight vasoconstriction sharply raises resistance.
- Cuff at heart level; a raised arm reads low.
- Laplace: surfactant keeps small alveoli open; a dilated vessel has higher wall tension.
- Ultrasound gel removes air; higher frequency: finer detail, less depth.
- Myopia: diverging lens; hyperopia, presbyopia: converging.
- Inspiration is Boyle's law: chest volume up, pressure down, air in.
- Gy measures energy absorbed; Sv weighs biological risk.

TRAP

- $N = mg$ only on a level surface with no other vertical force.
- A force perpendicular to motion does no work (centripetal, magnetic).
- No centrifugal force: net force points to the center.
- Faster fluid, **lower** pressure (Bernoulli).
- Blood pressure is a gauge pressure.
- Two equal sound sources add 3 dB, not double the level.
- Diverging lens alone: always virtual, upright, smaller.
- Capacitors in series add like resistors in parallel.
- Gas laws need kelvin.

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- Photoelectric: intensity sets the number of electrons, frequency their energy.
- Inelastic collision: momentum conserved, kinetic energy not.
- After two half-lives a quarter remains, not zero.

High-yield medical physics

Topic	Relation and worked value
MAP	MAP ≈ DBP + (SBP − DBP)/3: 120/80 mmHg ⇒ 93 mmHg. MAP − CVP ≈ CO × SVR, CO = HR × SV
Valve	$\Delta P \approx 4v^2$ (mmHg, m/s): v = 4 m/s ⇒ 64 mmHg. Severe aortic stenosis: $v_{max} \geq 4$ m/s
Ultrasound	c_{tissue} = 1,540 m/s; at 5 MHz, λ = 0.31 mm. Doppler $\Delta f = 2f_0 v\cos\theta/c$: v = 0.5 m/s, $\cos\theta$ = 0.5 ⇒ 1.6 kHz
Inspired O_2	$P_{IO_2} = F_{O_2}\left(P_{atm} - P_{H_2O}\right)$, P_{H_2O} = 47 mmHg: 0.21 × (760 − 47) ≈ 150 mmHg
Doses	Effective dose, adult: Chest X-ray (PA) 0.02 mSv; Chest X-ray (PA + lateral) 0.1 mSv; Mammography 0.4 mSv; Head CT 2 mSv; Chest CT 7 mSv; Abdomen CT 8 mSv; Natural background (per year) 3 mSv
Gy to Sv	w_R: photons (X, γ), electrons 1; protons 2; alpha particles 20

Constants (CODATA 2022)

c	2.998×10^8 m/s	k	8.988×10^9 N·m²/C²
h	6.626×10^{-34} J·s	ε_0	8.854×10^{-12} F/m
e	1.602×10^{-19} C	μ_0	1.257×10^{-6} N/A²
N_A	6.022×10^{23} mol⁻¹	m_e	9.109×10^{-31} kg
k_B	1.381×10^{-23} J/K	m_p	1.673×10^{-27} kg
R	8.314 J/(mol·K)	g_n	9.807 m/s²
R	0.08206 L·atm/(mol·K)	hc	1,240 eV·nm
G	6.674×10^{-11} N·m²/kg²		

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